

Residue Theory with Branch Points

Sections 8.6–8.7

Complex Analysis

Section 8.6: Integrands with Branch Points

Many integrals involve fractional powers such as

$$x^\alpha$$

Example:

$$\int_0^\infty \frac{x^\alpha}{1+x^2} dx$$

Key difficulty:

- ▶ z^α is multivalued
- ▶ we must choose a branch of the logarithm
- ▶ contour integrals must avoid the branch cut

Goal

We study integrals of the form

$$\int_0^\infty \frac{x^\alpha P(x)}{Q(x)} dx$$

where

- ▶ P, Q are polynomials
- ▶ $\deg Q \geq \deg P + 2$
- ▶ $0 < \alpha < 1$

The complex integrand is

$$f(z) = \frac{z^\alpha P(z)}{Q(z)}.$$

The Branch Cut

We define the branch

$$z^\alpha = e^{\alpha \log_0 z}$$

with

$$\log_0 z = \ln |z| + i \arg_0 z, \quad 0 < \arg_0 z < 2\pi.$$

Thus the positive real axis is a **branch cut**.

Approaching the positive real axis from above or below gives different arguments:

$$\arg z = \begin{cases} 0 & \text{(approaching from above)} \\ 2\pi & \text{(approaching from below)} \end{cases}$$

Contour Near the Positive Real Axis

The contour travels:

- ▶ from $x = r$ to $x = R$ along the **top** of the positive real axis
- ▶ around the large circle
- ▶ from $x = R$ back to $x = r$ along the **bottom** of the positive real axis
- ▶ around the small circle

Thus the positive real axis is traversed twice:

- ▶ once with argument 0
- ▶ once with argument 2π

L_1 : Integral Along the Top of the Branch Cut

Along the top of the positive real axis we set

$$z = x, \quad r < x < R$$

with

$$\arg z = 0.$$

Therefore

$$z^\alpha = x^\alpha.$$

The contribution of this segment is

$$\int_{L_1} f(z) dz = \int_r^R \frac{x^\alpha P(x)}{Q(x)} dx.$$

L_2 : Integral Along the Bottom of the Branch Cut

Along the lower side of the branch cut we again write

$$z = x$$

but now the argument is

$$\arg z = 2\pi.$$

Hence

$$z^\alpha = e^{\alpha(\ln x + i2\pi)} = x^\alpha e^{2\pi i\alpha}.$$

The path runs from R back to r , so

$$\int_{L_2} f(z) dz = \int_R^r \frac{x^\alpha e^{2\pi i\alpha} P(x)}{Q(x)} dx.$$

Finishing L_2 and adding the L_1 Integral

Reversing the limits gives

$$\int_R^r \frac{x^\alpha e^{2\pi i \alpha} P(x)}{Q(x)} dx = -e^{2\pi i \alpha} \int_r^R \frac{x^\alpha P(x)}{Q(x)} dx.$$

The contribution of both sides of the branch cut is

$$\int_r^R \frac{x^\alpha P(x)}{Q(x)} dx - e^{2\pi i \alpha} \int_r^R \frac{x^\alpha P(x)}{Q(x)} dx = (1 - e^{2\pi i \alpha}) \int_r^R \frac{x^\alpha P(x)}{Q(x)} dx.$$

Taking the Limits

Letting

$$r \rightarrow 0, \quad R \rightarrow \infty$$

and using the fact that the circular arcs (we are going to see in a minute) vanish, we obtain

$$(1 - e^{2\pi i \alpha}) \int_0^\infty \frac{x^\alpha P(x)}{Q(x)} dx = 2\pi i \sum \operatorname{Res}(f, z_k).$$

Therefore

$$\int_0^\infty \frac{x^\alpha P(x)}{Q(x)} dx = \frac{2\pi i}{1 - e^{2\pi i \alpha}} \sum \operatorname{Res}(f, z_k).$$

Keyhole Contour

We integrate around a contour consisting of

- ▶ the positive real axis
- ▶ a large circle C_R
- ▶ the negative real axis
- ▶ a small circle C_r

This contour avoids the branch cut along the positive real axis.

The contour encloses the poles of $P(z)/Q(z)$.

Theorem 8.6.1

Let

$$f(z) = \frac{z^\alpha P(z)}{Q(z)}, \quad 0 < \alpha < 1$$

with

$$\deg Q \geq \deg P + 2.$$

Then

$$\text{P.V.} \int_0^\infty \frac{x^\alpha P(x)}{Q(x)} dx = \frac{2\pi i}{1 - e^{2\pi i \alpha}} \sum \text{Res}(f, z_k)$$

where z_k are the nonzero poles of P/Q .

Proof Idea

We evaluate

$$\oint_C f(z) dz$$

using the residue theorem.

Since the poles inside C are $z_1, \dots, z_k$,

$$\oint_C f(z) dz = 2\pi i \sum \operatorname{Res}(f, z_k).$$

The main work is to compute the contributions of each part of the contour.

Decomposition of the Contour

Write

$$C = L_1 + C_R + L_2 + C_r$$

where

- ▶ L_1 : positive real axis
- ▶ L_2 : negative real axis
- ▶ C_R : large circle
- ▶ C_r : small circle

Thus

$$\oint_C f(z)dz = \int_{L_1} + \int_{C_R} + \int_{L_2} + \int_{C_r}.$$

Large Circle Contribution

On C_R

$$|f(z)| \approx \frac{|z|^\alpha |P(z)|}{|Q(z)|}.$$

Because

$$\deg Q \geq \deg P + 2$$

we have

$$f(z) = O(|z|^{\alpha-2}).$$

Thus

$$\int_{C_R} f(z) dz \rightarrow 0$$

as

$$R \rightarrow \infty.$$

Small Circle Contribution

On C_r ,

$$|f(z)| \approx |z|^\alpha.$$

Hence

$$\left| \int_{C_r} f(z) dz \right| \leq r^\alpha (2\pi r).$$

Since $\alpha > 0$,

$$\int_{C_r} f(z) dz \rightarrow 0$$

as

$$r \rightarrow 0.$$

Upper Real Axis

On the upper side of the branch cut,

$$z = x$$

with argument

$$\arg z = 0.$$

Thus

$$z^\alpha = x^\alpha.$$

The contribution is

$$\int_0^R \frac{x^\alpha P(x)}{Q(x)} dx.$$

Lower Real Axis

On the lower side,

$$\arg z = 2\pi.$$

Hence

$$z^\alpha = x^\alpha e^{2\pi i \alpha}.$$

The integral becomes

$$\int_R^0 \frac{x^\alpha e^{2\pi i \alpha} P(x)}{Q(x)} dx.$$

Combining the Real Axis Integrals

Adding the two segments gives

$$(1 - e^{2\pi i\alpha}) \int_0^\infty \frac{x^\alpha P(x)}{Q(x)} dx.$$

All other contour pieces vanish.

Therefore

$$(1 - e^{2\pi i\alpha}) \int_0^\infty \frac{x^\alpha P(x)}{Q(x)} dx = 2\pi i \sum \text{Res}(f, z_k).$$

Final Formula

Solving for the real integral gives

$$\int_0^\infty \frac{x^\alpha P(x)}{Q(x)} dx = \frac{2\pi i}{1 - e^{2\pi i \alpha}} \sum \text{Res}(f, z_k).$$

Done.

Section 8.7: Meromorphic Functions

A function f is **meromorphic** in a domain D if

- ▶ f is analytic except for isolated poles.
- ▶ Removable singularities are fine.
- ▶ Essential singularities are not.

Examples:

- ▶ rational functions
- ▶ $\tan z$
- ▶ $\frac{1}{\sin z}$

Argument Principle

If f is meromorphic inside a simple ccw closed contour C

$$\frac{1}{2\pi i} \int_C \frac{f'(z)}{f(z)} dz = Z - P$$

where

- ▶ Z = number of zeros inside C
- ▶ P = number of poles inside C

Argument Principle via Product Factorization

Let f be meromorphic inside and on a simple closed contour C with no zeros or poles on C .

Suppose inside C :

- ▶ $a_1, \dots, a_k$ are zeros with multiplicities $m_1, \dots, m_k$
- ▶ $b_1, \dots, b_\ell$ are poles with orders $n_1, \dots, n_\ell$

Then f can be written in the form

$$f(z) = \frac{(z - a_1)^{m_1} \cdots (z - a_k)^{m_k}}{(z - b_1)^{n_1} \cdots (z - b_\ell)^{n_\ell}} g(z)$$

where $g(z)$ is analytic and never zero in the region.

Logarithmic Derivative

Take the logarithmic derivative:

$$\frac{f'(z)}{f(z)} = \sum_{j=1}^k \frac{m_j}{z - a_j} - \sum_{j=1}^{\ell} \frac{n_j}{z - b_j} + \frac{g'(z)}{g(z)}.$$

Since $g(z) \neq 0$, the function $g'(z)/g(z)$ is analytic.

Integrating around C ,

$$\oint_C \frac{f'(z)}{f(z)} dz = \sum_{j=1}^k m_j \oint_C \frac{dz}{z - a_j} - \sum_{j=1}^{\ell} n_j \oint_C \frac{dz}{z - b_j}.$$

Each integral equals $2\pi i$, so

$$\oint_C \frac{f'(z)}{f(z)} dz = 2\pi i \left(\sum m_j - \sum n_j \right).$$

Thus

$$\frac{1}{2\pi i} \oint_C \frac{f'(z)}{f(z)} dz = Z - P.$$

Rouché's Theorem

Let f, g be analytic inside C .

If

$$|g(z)| < |f(z)| \quad z \in C$$

then

$$f \quad \text{and} \quad f + g$$

have the same number of zeros inside C .

This theorem is extremely useful for counting roots.

Summary

Section 8.6:

- ▶ integrals with branch points
- ▶ keyhole contour
- ▶ fractional powers z^α

Section 8.7:

- ▶ meromorphic functions
- ▶ argument principle
- ▶ Rouché's theorem